SORU KİTAPÇIĞI

Tarih: 15.06.2023

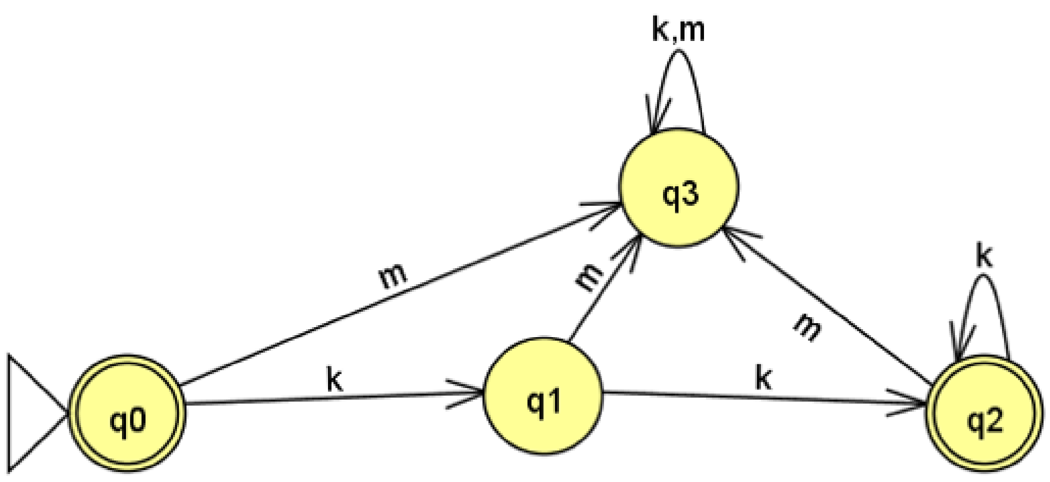
Sınavın,

Dersi:	BLM2502 - 1 Hesaplama Kuramı
Başlığı:	Final
Değerlendirme Türü:	Çoktan Seçmeli Sınav
Başlangıç Tarihi / Süresi:	6.06.2023 09:00 / 70 (Dk.)

Sorular,

1 / 25

What is the language of the automaton in the image.



A) $\{kk\}^*$
B) $\{kk\}^+$
C) kkk^*
D) $(kkk^*)^*$
E) None of the above

40,00 Puan

A A

B B

C C

D D

E E

2 / 25

Which one matches to strings with even number d's?

A) (c*dc*d)*
B) c*(dc*dc*)*
C) c*(dc*d*)*c*
D) c*d(dc*d)dc*

30,00 Puan

A A

B B

C C

D D

3 / 25

Consider the following languages.

$L_1 = \{0^p 1^q 0^r \mid p, q, r \geq 0\}$

$L_2 = \{0^p 1^q 0^r \mid p, q, r \geq 0$ and $p \neq q\}$

Which one of the following statements is FALSE?

A) L_1 is context-free.
B) L_2 is context-free.
C) L_1 intersection L_2 is context-free.
D) Complement of L_1 is context-free but not regular.
E) L_2 is subset of L_1.

40,00 Puan

A A

B B

C C

D D

E E

4 / 25

Which one is in Chomsky normal form?

A) $S \rightarrow TTTa|a$
$T \rightarrow c$

C) $S \rightarrow TT|a$
$T \rightarrow c$

B) $S \rightarrow TTAA|a$
$TA \rightarrow c$

D) $S \rightarrow Ta|a$
$T \rightarrow c$

30,00 Puan

A A

B B

C C

D D

5 / 25

Regular languages are finite.

A) True B) False

10,00 Puan

A A

B B

6 / 25

If K and L are regular, (K∪B)* is also regular.

A) True B) False

10,00 Puan

A A

B B

7 / 25

If T and M are context-free, $T \cap M$ is also context-free.

A) True B) False

10,00 Puan

A A

B B

8 / 25

$\{c^k d^k : 0 \leq k \leq 10\}$ is regular.

A) True B) False

10,00 Puan

A A

B B

9 / 25

$\{c^i d^i g^i : 0 \leq i\}$ is regular.

A) True B) False

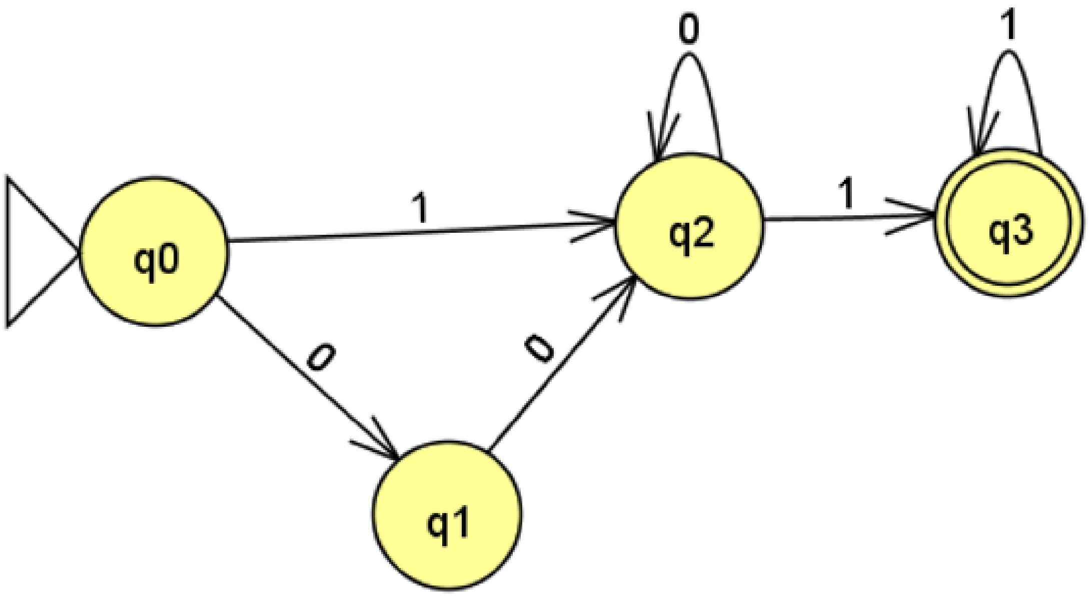
10,00 Puan

A A

B B

10 / 25

Let λ be empty string. Which of the following has the language of the automaton in the image?



A) $(00 \cup \lambda)\, 0^{*}11^{*}$

B) $(000^{*} \cup 10^{*})\, 0^{*}11^{*}$

C) $(00 \cup 1)\, 0^{*}11^{*}$

D) $(10^{*} \cup \lambda)\, 0^{*}11^{*}$

E) $(10^{*} \cup 0^{*}1)\, 0^{*}11^{*}$

40,00 Puan

A A

B B

C C

D D

E E

11 / 25

$\{a^m b^n (ac)^{m+n} \mid m, n \geq 1\}$ is context-free.

A) True B) False

10,00 Puan

A A

B B

12 / 25

Concatenation of two regular languages is context-free.

A) True B) False

10,00 Puan

A A

B B

13 / 25

Every context free language has a TM.

A) True B) False

10,00 Puan

A A

B B

14 / 25

TMs are non-deterministic but more powerful than PDAs.

A) True B) False

10,00 Puan

A A

B B

15 / 25

A regular language is not a CFL if it has more than one parsing tree at least for one string of its language.

10,00 Puan

A A

B B

16 / 25

A language L has a CFG if and only if a TM recognizes the language L.

A) True B) False

10,00 Puan

A A

B B

17 / 25

It can be proven that computers cannot solve all problems.

A) True B) False

10,00 Puan

A A

B B

18 / 25

Following CFG generates $\{0^m21^m0^n12^n \mid m \geq 1,\ n \geq 1\}$.

S → 0X10Y2

X → 0X1 | 2

Y → 0Y2 | 1

A) True B) False

10,00 Puan

A A

B B

19 / 25

PDAs can simulate machines that accept the intersection of CFLs.

A) True B) False

10,00 Puan

A A

B B

20 / 25

Which one of the following is FALSE for a Turing machine M that accepts the language L.

A) L is Turing Recognizable.
B) M may loop forever for some strings not in L
C) M loops forever for all strings in L
D) None of them.

10,00 Puan

A A

B B

C C

D D

21 / 25

Every non-deterministic Turing machine has an equivalent multi-tape Turing machine.

A) True B) False

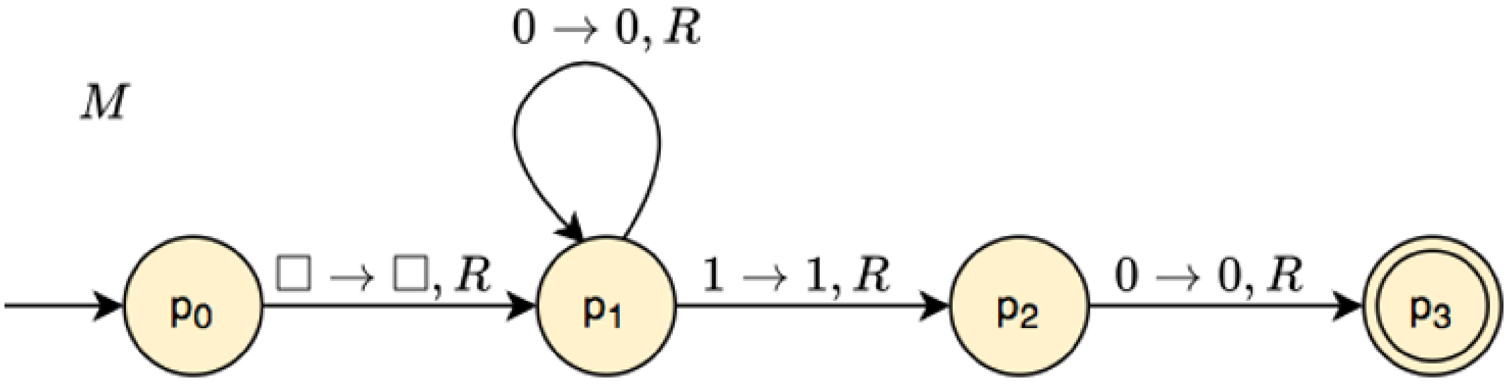
10,00 Puan

A A

B B

22 / 25

Which has the same language as the following TM?



A) 0^*10
B) 01^*0
C) $0^*10(1 \cup 0)^*$
D) $01^*0(1 \cup 0)^*$
E) None of the above.

10,00 Puan

A A

B B

C C

D D

E E

23 / 25

Turing machine variants that have tapes with only one infinite side have less power than the Standard Turing Machines that have tapes with two infinite sides.

A) True B) False

10,00 Puan

A A

B B

24 / 25

When we want to prove that two Turing Machine variants have the same power, we show that machines of first variant simulate machines of second variant. There is no need to prove that machines of second variant simulate machines of first variant.

A) True B) False

10,00 Puan

A A

B B

25 / 25

If a problem is solvable by an algorithm, then there is a Turing Machine that can execute that algorithm.

A) True B) False

10,00 Puan

A A

B B

Your points from this exam will be calculated using the formula min(120(p/e)-20(t-a)/(z-a), 100), where e is the maximum possible raw points one can get from the exam, p is your raw points, a is the start time of the exam, z is the end time of the exam, and t is the time you finish the exam. In some automaton drawings a triangle shows the initial state. DFA: Deterministic Finite Automaton, NFA: Non-deterministic Finite Automaton, PDA: Push Down Automata, DPDA: Deterministic Push Down Automaton, NPDA: Non-deterministic Push Down Automaton, CFG: Context-Free Grammar, CFL: Context-Free Language, TM: Turing Machine, STM: Standard Turing Machine.